

Round 4 — search-enabled P_5 vs no-search baseline
25 paired cars

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Per-car summary — 25 paired cars (no-search vs search-enabled P5)

Same 25 VINs run twice: once in rounds 1+2 with no-search P5 (`smoke_60_v5a_merged` + `smoke_90_v5a_round2`), and once in round 4 with search-enabled P5 (`smoke_25_v5b_search_round4`). P_1 – P_3 are deterministic so we report them once. P_4 (GPT-5.5 DB-tool agent) is non-deterministic so both runs are shown. N_s = number of web searches the search-enabled P_5 issued.

Car	B	Truth	P ₁	P ₂	P ₃	No-search baseline					N _s	Search-enabled (v5b)			
						P ₄	P ₅	D end	I end	P ₄		P ₅	D end	I end	
1985 Toyota Pickup	N	\$18,888	n/a	\$19,070	\$29,321	\$28,000	\$26,000	\$21,000	\$21,000	1	\$27,500	\$26,000	\$13,000	\$19,000	
1986 Porsche 911	N	\$51,000	\$84,518	\$96,691	\$75,065	\$67,500	\$75,000	\$78,000	\$79,500	1	\$88,000	\$79,000	\$84,000	\$79,000	
1987 Land Rover Defender	N	\$32,000	\$46,250	\$41,814	\$37,936	\$52,000	\$44,200	\$44,200	\$44,200	1	\$55,000	\$44,500	\$44,000	\$38,750	
1990 Nissan Z-car	N	\$9,600	\$22,251	\$14,837	\$11,309	\$12,500	\$13,200	\$12,700	\$13,500	1	\$15,000	\$14,500	\$15,500	\$13,250	
1995 Ford F-Series	N	\$26,500	\$17,669	\$31,889	\$31,444	\$42,500	\$33,000	\$29,800	\$23,500	1	\$34,000	\$32,500	\$32,500	\$32,500	
1995 Hummer H1	N	\$43,500	n/a	\$100,977	\$63,442	\$108,000	\$92,000	\$84,000	\$77,500	1	\$80,000	\$86,000	\$74,750	\$86,000	
1997 Land Rover Defender	N	\$26,000	\$55,477	\$37,600	\$35,774	\$62,000	\$53,000	\$33,000	\$45,000	1	\$50,000	\$50,000	\$44,100	\$39,000	
2004 Mercedes-B SL-Class	N	\$12,500	\$15,797	\$17,015	\$18,359	\$15,500	\$16,800	\$16,950	\$16,000	1	\$16,000	\$16,650	\$16,650	\$16,650	
2008 BMW 3 Series	N	\$21,000	\$22,831	\$28,038	\$21,433	\$32,000	\$28,500	\$25,200	\$25,750	1	\$23,000	\$23,000	\$22,500	\$21,500	
2008 Mercedes-B CLK	N	\$13,777	\$31,534	\$15,715	\$15,118	\$18,500	\$16,800	\$17,500	\$16,600	1	\$16,500	\$16,400	\$16,500	\$16,400	
2014 Ford F-Series	N	\$21,500	\$36,262	\$26,575	\$26,123	\$23,500	\$25,800	\$26,600	\$26,200	1	\$25,500	\$26,500	\$26,000	\$27,500	
2016 Porsche 911	N	\$90,500	\$155,044	\$115,300	\$104,520	\$112,000	\$112,000	\$112,000	\$112,000	1	\$110,000	\$114,000	\$112,000	\$110,000	
2017 Ford F-Series	N	\$40,000	\$49,060	\$46,205	\$44,443	\$47,000	\$47,000	\$47,000	\$44,500	1	\$44,000	\$46,500	\$46,300	\$46,500	
2019 Fiat 124	N	\$20,500	n/a	\$16,795	\$24,227	\$27,000	\$25,500	\$23,500	\$24,600	1	\$27,000	\$24,500	\$24,300	\$24,750	
2023 Audi R8	N	\$159,770	\$254,500	\$154,050	\$155,382	\$225,000	\$235,000	\$231,000	\$223,000	1	\$225,000	\$220,000	\$212,000	\$185,000	
1984 Mercedes-B SL-Class	S	\$26,000	\$25,598	\$30,925	\$19,359	\$32,000	\$23,500	\$24,500	\$23,500	1	\$27,000	\$27,500	\$26,500	\$27,500	
1995 Jeep Wrangler	S	\$20,500	\$14,350	\$22,555	\$24,216	\$28,000	\$25,500	\$22,250	\$23,500	1	\$25,000	\$24,000	\$24,000	\$23,000	
2004 Mercedes-B CLK	S	\$9,600	\$16,261	\$17,805	\$15,930	\$21,000	\$17,800	\$17,600	\$17,500	1	\$21,000	\$16,800	\$16,100	\$17,400	
2004 Porsche Boxster	S	\$25,650	\$23,035	\$18,958	\$19,066	\$20,000	\$20,100	\$18,800	\$20,400	1	\$20,500	\$20,800	\$20,000	\$21,200	
2004 Porsche Boxster	S	\$18,550	\$16,750	\$17,224	\$17,332	\$18,500	\$17,850	\$17,750	\$17,450	1	\$19,000	\$18,750	\$17,275	\$18,750	
2005 Chrysler Crossfire	S	\$18,500	\$15,111	\$12,552	\$12,941	\$16,500	\$15,500	\$14,000	\$15,500	2	\$19,000	\$16,800	\$13,200	\$15,800	
2007 Hummer H2	S	\$46,200	\$23,100	\$29,575	\$36,481	\$30,000	\$32,000	\$28,800	\$27,000	1	\$30,000	\$29,000	\$28,000	\$25,500	
2008 Jeep Wrangler	S	\$19,500	\$19,107	\$31,856	\$22,831	\$25,000	\$23,500	\$21,750	\$24,000	1	\$30,000	\$21,500	\$21,500	\$21,000	
2010 Mercedes-B CL	S	\$46,200	n/a	\$44,425	\$40,934	\$50,000	\$47,000	\$44,000	\$44,000	1	\$45,000	\$47,500	\$45,000	\$45,000	
2012 Bentley Continental G	S	\$46,200	\$71,580	\$66,510	\$56,305	\$62,000	\$65,000	\$64,300	\$65,000	1	\$86,000	\$70,000	\$56,500	\$68,000	

B = bucket (S = sold, N = not_sold). Truth = realized sale price (S) or high_bid (N). P_1 – P_3 reproduced once because they are deterministic (local DB averaging, k-NN, and the cluster-trained moe_D model respectively).

Per-signal MAE / MAPE / RMSE — pooled headline

Signal	N	MAE	MAPE	RMSE
P5 (no search, baseline)	25	\$12,780	35.2%	\$20,900
P5 (search, v5b)	25	\$11,850	32.3%	\$18,638
D end (no search)	25	\$11,150	28.8%	\$19,016
D end (search)	25	\$10,530	30.0%	\$16,116
I end (no search)	25	\$11,170	30.2%	\$17,811
I end (search)	25	\$9,324	27.1%	\$14,135

Search-usage summary

Distribution of how many web searches the new P5 actually issued for the 25 valid cars.

n_searches issued	N cars
1	24
2	1
mean	1.04
median	1
max	2

Round 4 (search-enabled) — standalone aggregate analysis

Iteration counts (D vs I, paired, n=25)

D mean = 5.32 (sd 2.56), I mean = 4.28 (sd 2.17). Differences (D-I): mean = 1.04 (sd 2.67), range [-6, 7].

Statistical validity — paired tests on H_0 : mean(D-I) = 0

Test	Statistic	p-value
Shapiro-Wilk on diffs (normality)	$W = 0.9223$	0.058
Paired t -test (df=24)	$t = 1.948$	0.063
Wilcoxon signed-rank	$W = 35.5$	0.051
Bootstrap (10,000 paired resamples)	95% CI for mean diff: [0.00, 2.04]	0.053
Cohen's d_z (effect size, paired)	0.390 (small)	—

Per-signal MAE / MAPE / RMSE — pooled (n=25)

Signal	N	MAE	MAPE	RMSE
P1	21	\$17,838	48.6%	\$28,854
P2	25	\$10,984	34.9%	\$17,321
P3	25	\$6,964	24.5%	\$8,854
P4	25	\$13,606	39.1%	\$20,818
P5	25	\$11,850	32.3%	\$18,638
D end	25	\$10,530	30.0%	\$16,116
I end	25	\$9,324	27.1%	\$14,135

Per-signal MAE / MAPE / RMSE — split by outcome

SOLD = realized sale price; NOT_SOLD = high_bid (lower bound on market value).

Signal	N	sold=1 (n=10)			N	sold=0 (n=15)		
		MAE	MAPE	RMSE		MAE	MAPE	RMSE
P1	9	\$7,765	27.3%	\$11,934	12	\$25,392	64.6%	\$36,745
P2	10	\$8,021	32.7%	\$10,111	15	\$12,959	36.3%	\$20,781
P3	10	\$5,846	24.3%	\$6,396	15	\$7,709	24.6%	\$10,168
P4	10	\$9,070	34.7%	\$14,616	15	\$16,631	42.1%	\$24,081
P5	10	\$6,325	22.9%	\$9,801	15	\$15,534	38.7%	\$22,692
D end	10	\$5,442	21.9%	\$7,477	15	\$13,922	35.4%	\$19,889
I end	10	\$6,435	23.5%	\$10,019	15	\$11,251	29.5%	\$16,312

NOT_SOLD — delta analysis (n=15)

δ = high_bid - prediction. Positive δ = prediction below high_bid = **recovered**; negative δ = prediction above high_bid = **not recovered**.

Signal	N	MAE	MAPE	$\delta > 0$ (recovered)			$\delta < 0$ (not recovered)		
				N ₊	%	mean δ_+	N ₋	%	mean δ_-
P1	12	\$25,392	64.6%	1	8%	\$8,830	11	92%	\$-26,898
P2	15	\$12,959	36.3%	2	13%	\$4,712	13	87%	\$-14,228
P3	15	\$7,709	24.6%	1	7%	\$4,387	14	93%	\$-7,946
P4	15	\$16,631	42.1%	0	0%	n/a	15	100%	\$-16,631
P5	15	\$15,534	38.7%	0	0%	n/a	15	100%	\$-15,534
D end	15	\$13,922	35.4%	1	7%	\$5,888	14	93%	\$-14,496
I end	15	\$11,251	29.5%	0	0%	n/a	15	100%	\$-11,251

True market value sits between high_bid and the (failed) seller reserve, so a small negative δ may still reflect a reasonable reserve; large negative δ means the system overshot. Positive δ definitely indicates the system would have salvaged the sale.

Round 4 baseline (no search, same 25 VINs from rounds 1+2) — standalone aggregate analysis

Iteration counts (D vs I, paired, n=25)

D mean = 5.72 (sd 2.73), I mean = 4.52 (sd 2.29). Differences (D-I): mean = 1.20 (sd 2.68), range [-5, 6].

Statistical validity — paired tests on H_0 : mean(D-I) = 0

Test	Statistic	p-value
Shapiro-Wilk on diffs (normality)	$W = 0.9131$	0.036
Paired t -test (df=24)	$t = 2.241$	0.035
Wilcoxon signed-rank	$W = 28.0$	0.037
Bootstrap (10,000 paired resamples)	95% CI for mean diff: [0.20, 2.24]	0.024
Cohen's d_z (effect size, paired)	0.448 (small)	—

Per-signal MAE / MAPE / RMSE — pooled (n=25)

Signal	N	MAE	MAPE	RMSE
P1	21	\$17,838	48.6%	\$28,854
P2	25	\$10,984	34.9%	\$17,321
P3	25	\$6,964	24.5%	\$8,854
P4	25	\$14,394	42.9%	\$22,141
P5	25	\$12,780	35.2%	\$20,900
D end	25	\$11,150	28.8%	\$19,016
I end	25	\$11,170	30.2%	\$17,811

Per-signal MAE / MAPE / RMSE — split by outcome

SOLD = realized sale price; NOT_SOLD = high_bid (lower bound on market value).

Signal	sold=1 (n=10)				sold=0 (n=15)			
	N	MAE	MAPE	RMSE	N	MAE	MAPE	RMSE
P1	9	\$7,765	27.3%	\$11,934	12	\$25,392	64.6%	\$36,745
P2	10	\$8,021	32.7%	\$10,111	15	\$12,959	36.3%	\$20,781
P3	10	\$5,846	24.3%	\$6,396	15	\$7,709	24.6%	\$10,168
P4	10	\$7,390	31.7%	\$9,027	15	\$19,064	50.3%	\$27,617
P5	10	\$6,275	25.5%	\$8,429	15	\$17,117	41.7%	\$26,089
D end	10	\$6,335	24.6%	\$8,817	15	\$14,361	31.6%	\$23,471
I end	10	\$6,745	25.9%	\$9,287	15	\$14,121	33.1%	\$21,708

NOT_SOLD — delta analysis (n=15)

δ = high_bid - prediction. Positive δ = prediction below high_bid = **recovered**; negative δ = prediction above high_bid = **not recovered**.

Signal	N	MAE	MAPE	$\delta > 0$ (recovered)			$\delta < 0$ (not recovered)		
				N ₊	%	mean δ_+	N ₋	%	mean δ_-
P1	12	\$25,392	64.6%	1	8%	\$8,830	11	92%	\$-26,898
P2	15	\$12,959	36.3%	2	13%	\$4,712	13	87%	\$-14,228
P3	15	\$7,709	24.6%	1	7%	\$4,387	14	93%	\$-7,946
P4	15	\$19,064	50.3%	0	0%	n/a	15	100%	\$-19,064
P5	15	\$17,117	41.7%	0	0%	n/a	15	100%	\$-17,117
D end	15	\$14,361	31.6%	0	0%	n/a	15	100%	\$-14,361
I end	15	\$14,121	33.1%	1	7%	\$3,000	14	93%	\$-14,915

True market value sits between high_bid and the (failed) seller reserve, so a small negative δ may still reflect a reasonable reserve; large negative δ means the system overshot. Positive δ definitely indicates the system would have salvaged the sale.

Paired statistical tests — search vs no-search (25 cars: 10 SOLD + 15 NOT_SOLD)

For each car, both configurations were run. Tests are paired on per-car $|\text{error}|$ difference. Negative Δ = search-enabled is closer to truth.

Caveat: $n=25$ is small. None of these paired tests are powered to detect the $\sim \$1000$ - $\$2000$ magnitude differences observed in the descriptive tables. They are reported for completeness; treat them as exploratory.

Paired error comparison: P5 (no search) vs P5 (search) (n=25)

P5 (no search) MAE=\$12,781 MAPE=35.2% RMSE=\$20,900

P5 (search) MAE=\$11,851 MAPE=32.3% RMSE=\$18,639

Δ (P5 (search) – P5 (no search)) per-car $|\text{err}|$: mean = \$-930, sd = \$3,805; negative Δ = search helps.

Test (H_0 : Δ P5 (search)–P5 (no search) = 0)	Statistic	p-value
Shapiro–Wilk on Δ (normality)	$W = 0.7967$	0.0002
Paired t -test (df=24)	$t = -1.222$	0.234
Wilcoxon signed-rank	$W = 105.5$	0.203
Bootstrap (10,000 paired resamples)	95% CI for mean Δ : [\$-2,484, \$+376]	—
Cohen's d_z (effect size, paired)	-0.244 (small)	—

Paired error comparison: D end (no search) vs D end (search) (n=25)

D end (no search) MAE=\$11,151 MAPE=28.8% RMSE=\$19,017

D end (search) MAE=\$10,531 MAPE=30.0% RMSE=\$16,116

Δ (D end (search) – D end (no search)) per-car $|\text{err}|$: mean = \$-620, sd = \$5,426; negative Δ = search helps.

Test (H_0 : Δ D end (search)–D end (no search) = 0)	Statistic	p-value
Shapiro–Wilk on Δ (normality)	$W = 0.8189$	0.0005
Paired t -test (df=24)	$t = -0.571$	0.573
Wilcoxon signed-rank	$W = 139.5$	0.764
Bootstrap (10,000 paired resamples)	95% CI for mean Δ : [\$-2,883, \$+1,317]	—
Cohen's d_z (effect size, paired)	-0.114 (small)	—

Paired error comparison: I end (no search) vs I end (search) (n=25)

I end (no search) MAE=\$11,171 MAPE=30.2% RMSE=\$17,812

I end (search) MAE=\$9,325 MAPE=27.1% RMSE=\$14,136

Δ (I end (search) – I end (no search)) per-car $|\text{err}|$: mean = \$-1,846, sd = \$8,068; negative Δ = search helps.

Test (H_0 : Δ I end (search)–I end (no search) = 0)	Statistic	p-value
Shapiro–Wilk on Δ (normality)	$W = 0.5298$	$7.46e - 08$
Paired t -test (df=24)	$t = -1.144$	0.264
Wilcoxon signed-rank	$W = 115.0$	0.201
Bootstrap (10,000 paired resamples)	95% CI for mean Δ : [\$-5,338, \$+522]	—
Cohen's d_z (effect size, paired)	-0.229 (small)	—

Paired statistical tests — split by outcome

SOLD (n=10)

Paired error comparison: P5 (no search) vs P5 (search) (n=10)

P5 (no search) MAE=\$6,275 MAPE=25.5% RMSE=\$8,429

P5 (search) MAE=\$6,325 MAPE=22.9% RMSE=\$9,802

Δ (P5 (search) – P5 (no search)) per-car |err|: mean = \$+50, sd = \$2,234; negative Δ = search helps.

Test (H_0 : Δ P5 (search)–P5 (no search) = 0)	Statistic	p-value
Shapiro–Wilk on Δ (normality)	$W = 0.7899$	0.011
Paired t -test (df=9)	$t = 0.071$	0.945
Wilcoxon signed-rank	$W = 20.5$	0.510
Bootstrap (10,000 paired resamples)	95% CI for mean Δ : [\$-1,100, \$+1,530]	—
Cohen's d_z (effect size, paired)	0.022 (small)	—

Paired error comparison: D end (no search) vs D end (search) (n=10)

D end (no search) MAE=\$6,335 MAPE=24.6% RMSE=\$8,817

D end (search) MAE=\$5,442 MAPE=21.9% RMSE=\$7,478

Δ (D end (search) – D end (no search)) per-car |err|: mean = \$-892, sd = \$2,650; negative Δ = search helps.

Test (H_0 : Δ D end (search)–D end (no search) = 0)	Statistic	p-value
Shapiro–Wilk on Δ (normality)	$W = 0.7480$	0.0034
Paired t -test (df=9)	$t = -1.065$	0.315
Wilcoxon signed-rank	$W = 18.0$	0.369
Bootstrap (10,000 paired resamples)	95% CI for mean Δ : [\$-2,648, \$+388]	—
Cohen's d_z (effect size, paired)	-0.337 (small)	—

Paired error comparison: I end (no search) vs I end (search) (n=10)

I end (no search) MAE=\$6,745 MAPE=25.9% RMSE=\$9,288

I end (search) MAE=\$6,435 MAPE=23.5% RMSE=\$10,019

Δ (I end (search) – I end (no search)) per-car |err|: mean = \$-310, sd = \$1,602; negative Δ = search helps.

Test (H_0 : Δ I end (search)–I end (no search) = 0)	Statistic	p-value
Shapiro–Wilk on Δ (normality)	$W = 0.8940$	0.188
Paired t -test (df=9)	$t = -0.612$	0.556
Wilcoxon signed-rank	$W = 17.5$	0.338
Bootstrap (10,000 paired resamples)	95% CI for mean Δ : [\$-1,180, \$+680]	—
Cohen's d_z (effect size, paired)	-0.194 (small)	—

NOT_SOLD (n=15)

Paired error comparison: P5 (no search) vs P5 (search) (n=15)

P5 (no search) MAE=\$17,118 MAPE=41.7% RMSE=\$26,090

P5 (search) MAE=\$15,534 MAPE=38.7% RMSE=\$22,693

Δ (P5 (search) – P5 (no search)) per-car |err|: mean = \$-1,583, sd = \$4,525; negative Δ = search helps.

Test (H_0 : Δ P5 (search)–P5 (no search) = 0)	Statistic	p-value
Shapiro–Wilk on Δ (normality)	$W = 0.7909$	0.0028
Paired t -test (df=14)	$t = -1.355$	0.197
Wilcoxon signed-rank	$W = 36.0$	0.300
Bootstrap (10,000 paired resamples)	95% CI for mean Δ : [\$-3,970, \$+347]	—
Cohen's d_z (effect size, paired)	-0.350 (small)	—

Paired error comparison: D end (no search) vs D end (search) (n=15)

D end (no search) MAE=\$14,361 MAPE=31.6% RMSE=\$23,471

D end (search) MAE=\$13,923 MAPE=35.4% RMSE=\$19,890

 Δ (D end (search) – D end (no search)) per-car $|\mathbf{err}|$: mean = \$-438, sd = \$6,773; negative Δ = search helps.

Test (H_0 : Δ D end (search)–D end (no search) = 0)	Statistic	p-value
Shapiro–Wilk on Δ (normality)	$W = 0.8578$	0.022
Paired t -test (df=14)	$t = -0.251$	0.806
Wilcoxon signed-rank	$W = 49.5$	0.851
Bootstrap (10,000 paired resamples)	95% CI for mean Δ : [\$-4,012, \$+2,613]	—
Cohen’s d_z (effect size, paired)	-0.065 (small)	—

Paired error comparison: I end (no search) vs I end (search) (n=15)

I end (no search) MAE=\$14,121 MAPE=33.1% RMSE=\$21,709

I end (search) MAE=\$11,251 MAPE=29.5% RMSE=\$16,313

 Δ (I end (search) – I end (no search)) per-car $|\mathbf{err}|$: mean = \$-2,870, sd = \$10,351; negative Δ = search helps.

Test (H_0 : Δ I end (search)–I end (no search) = 0)	Statistic	p-value
Shapiro–Wilk on Δ (normality)	$W = 0.6155$	$3.54e - 05$
Paired t -test (df=14)	$t = -1.074$	0.301
Wilcoxon signed-rank	$W = 44.0$	0.363
Bootstrap (10,000 paired resamples)	95% CI for mean Δ : [\$-8,690, \$+1,020]	—
Cohen’s d_z (effect size, paired)	-0.277 (small)	—